

Client Overview

Client	Large Construction Company in India
Location	Chennai, Tamil Nadu
Facility Type	Corporate Technology Center
Scale	17 Floors (3 Basements + Ground + 13 Upper Floors)
Objective	Enable Intelligent Energy & Environmental Optimization from Day One



Business Objectives

- Establish structured energy baselines from inception
- Optimize HVAC and environmental conditions
- Ensure regulatory and ESG alignment
- Enable granular visibility across floors and equipment
- Create a scalable digital foundation for future intelligence layers

Smart Energy & Environmental Optimization for a 17-Floor Digital Facility

Key Challenges

► Limited Asset Visibility

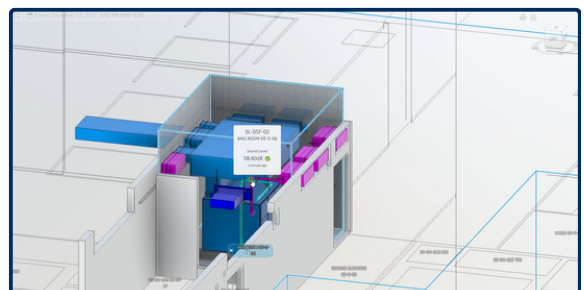
- Difficulty locating critical assets across the facility
- Maintenance schedules are not centrally managed
- Historical maintenance records are difficult to retrieve

► Environmental & HVAC Optimization

- Balancing comfort and energy reduction
- Managing AQI, temperature, humidity, occupancy
- Ensuring consistent performance across 17 floors

► Limited Operational Visibility

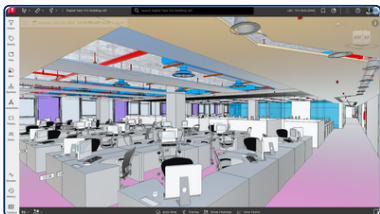
- BMS provided control, but not deep analytics
- Need for equipment-level intelligence
- Gap between control systems & actionable insights



Solution : A Unified Digital Twin for Environment and Energy Management (EEMS) ▼

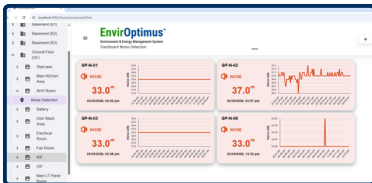
Integrated Deployment of :
(EEMS - EnviroOptimus Layer)

- BMS Integration
- IoT Sensors & Environmental Monitoring
- Data Analytics & Optimization Engine
- Digital Twin–Ready Foundation



Systems and Infrastructure
Deployed :

- 281 Smart Sensors Installed
- 2000+ BMS Points Integrated
- Floor-wise and zone-wise energy monitoring
- Environmental & utility load monitoring with forecasting



The Intelligent Optimization Layer for Smarter, Sustainable Operations ▼

EnviroOptimus acted as the optimization engine, moving beyond control into intelligence, transforming traditional Building Management Systems from control-centric operations into intelligent, data-driven decision-making. Optimize energy and environmental performance, verify asset health, enable predictive maintenance, and support occupant comfort, sustainability, and WELL Building initiatives.

Delivered Capabilities:



Granular Sub-Metering & Load Profiling



Energy Intensity Tracking (kWh/sq.ft.)



HVAC Performance Analytics



Environmental Monitoring Integration



Baseline Creation for a New Facility



Early Anomaly Detection



Peak Demand Optimization

Digital Twin Enablement for Asset Location and Maintenance ▼

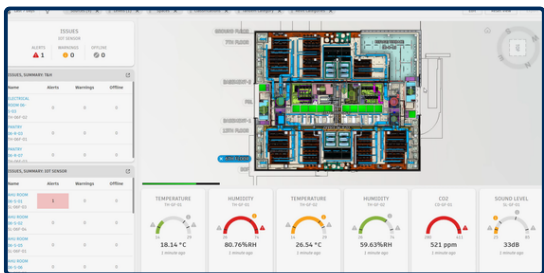
As part of the digital maturity roadmap:

- Asset rich BIM models form the robust foundation
- Operational & energy data structured through EEMS
- Clean, connected, and standardized data streams



Digital Twin readiness ensures:

- Asset-level data continuity
- Lifecycle intelligence
- Helps in “What if Analysis”



This positions the facility for predictive simulation and long-term autonomous optimization.



▶ **Smart Building from Day One**

Structured energy baselines were established immediately upon deployment, giving the facility a data-driven foundation.

▶ **Improved HVAC & Indoor Environmental Quality**

Data-driven thermal and air quality optimization delivered measurable improvements in occupant comfort and IEQ.

▶ **Energy Cost and Demand Optimization**

Load balancing and peak demand control reduced energy costs and eliminated avoidable demand charges.

▶ **ESG & Sustainability Alignment**

Carbon tracking and environmental performance reporting strengthened ESG compliance and sustainability disclosures.

▶ **Granular Decision Intelligence**

Equipment and zone-level analytics enabled continuous, data-backed operational improvements.

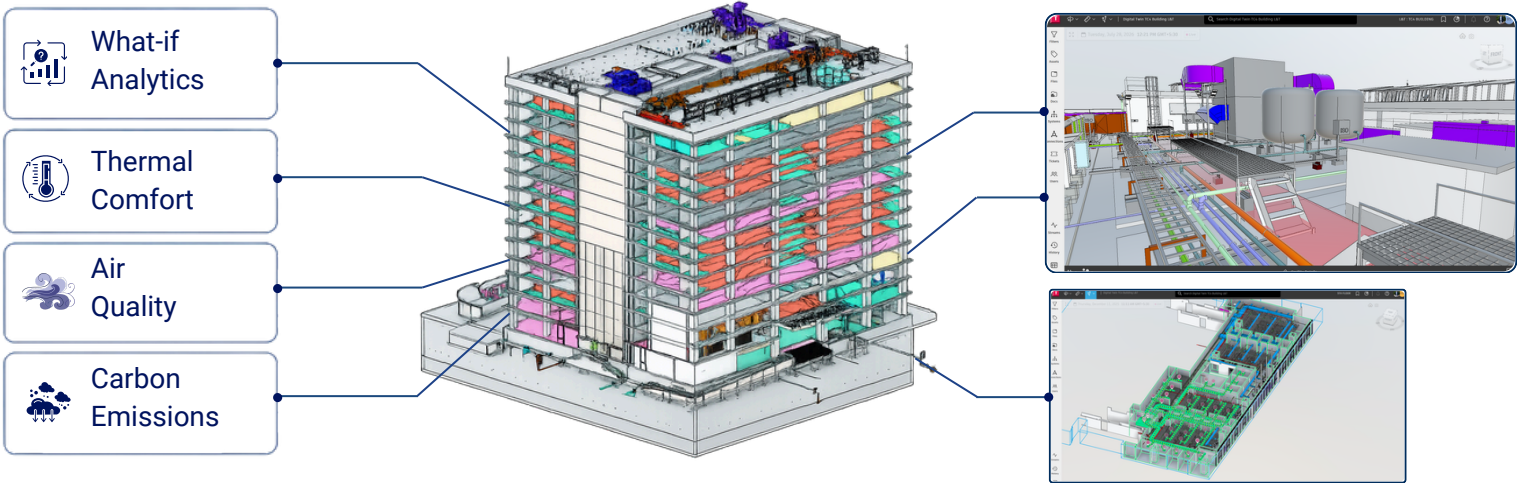
▶ **Scalable Digital Foundation**

A future-ready platform was established, laying the groundwork for predictive analytics and full digital twin capabilities.

Conclusion



With the deployment of EnviroOptimus and a digital twin-ready architecture, facilities start to fill up as floors become operational and transformed into smart, sustainable, and intelligence-driven assets.



From real-time environmental visibility to structured energy optimization, the project demonstrates how a new building can leap directly into a high-maturity digital ecosystem without waiting years to accumulate the data typically needed to get there.

EnviroOptimus : From Data to Performance

Extending BMS beyond control into deep analytics, environmental intelligence, and future Digital Twin enablement.

